

1. (20 marks) For $(a, b) \in \mathbb{R}^2$, consider the following quadratic form

$$Q(\mathbf{x}) = b(x_1^2 + x_2^2 + x_3^2) + 2a(x_1x_2 + x_2x_3).$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is, find the symmetric matrix A satisfying $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$. Then, find all the principal minors of A .
- (b) (12 marks) Define what it means for A to be *indefinite*, *positive semidefinite* and *negative semidefinite*. Then, find all parameter values for a such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*?
- (c) (4 marks) Plot the pairs (a, b) for which $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite* in $\mathbb{R}^2$.
2. (20 marks) Consider the functions $f(\mathbf{x}) = -x_1^2 - x_2^2$ defined on some convex set U in $\mathbb{R}^2$.
- (a) (6 marks) Define what it means for f to be *quasi-concave* and *quasi-convex*.
- (b) (14 marks) Is f *quasi-concave* or *quasi-convex*? Hint: Draw level sets to gain intuition and use counterexamples to show that a property fails for some U .
3. (40 marks) For $a > 0$, consider the problem:

$$\max_{x,y} \sqrt{xy} \quad \text{s.t.} \quad ax + y \leq 16, \quad x \geq a, \quad y \geq 0.$$

- (a) (10 marks) Is the objective function concave? Are the constraints convex? For what values of $a > 0$ is Slater condition satisfied? So, for what values of $a > 0$, can you solve this optimization problem using the *Kuhn-Tucker method*?
- (b) (6 marks) Suppose the restrictions on a found in part (a) hold. Write down the *Kuhn-Tucker necessary first order conditions* that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian?
- (c) (18 marks) Solve the constrained optimization problem for all values of $a > 0$ satisfying the restrictions on a found in part (a).
- (d) (6 marks) Suppose the restrictions on a found in part (a) hold. Use the Envelope Theorem to express the slope of value function with respect to the parameter a in terms of the solution $(x(a), y(a), \boldsymbol{\lambda}(a))$ found in part (c).
4. (20 Marks) Consider the correspondence $\mathbf{h} : [0, 4]^2 \rightrightarrows [0, 4]^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$ and

$$h_1(\mathbf{x}) = \begin{cases} [0, 2] & \text{if } x_2 = 0 \\ [1, 2] & \text{if } x_2 \in (0, 4) \\ [1, 4] & \text{if } x_2 = 4 \end{cases} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{cases} [0, 3] & \text{if } x_1 = 0 \\ [2, 3] & \text{if } x_1 \in (0, 4) \\ [2, 4] & \text{if } x_1 = 4 \end{cases}.$$

Argue why *Kakutani fixed point theorem* applies. Then, find all the *fixed points* of this correspondence.